

Cryptography II

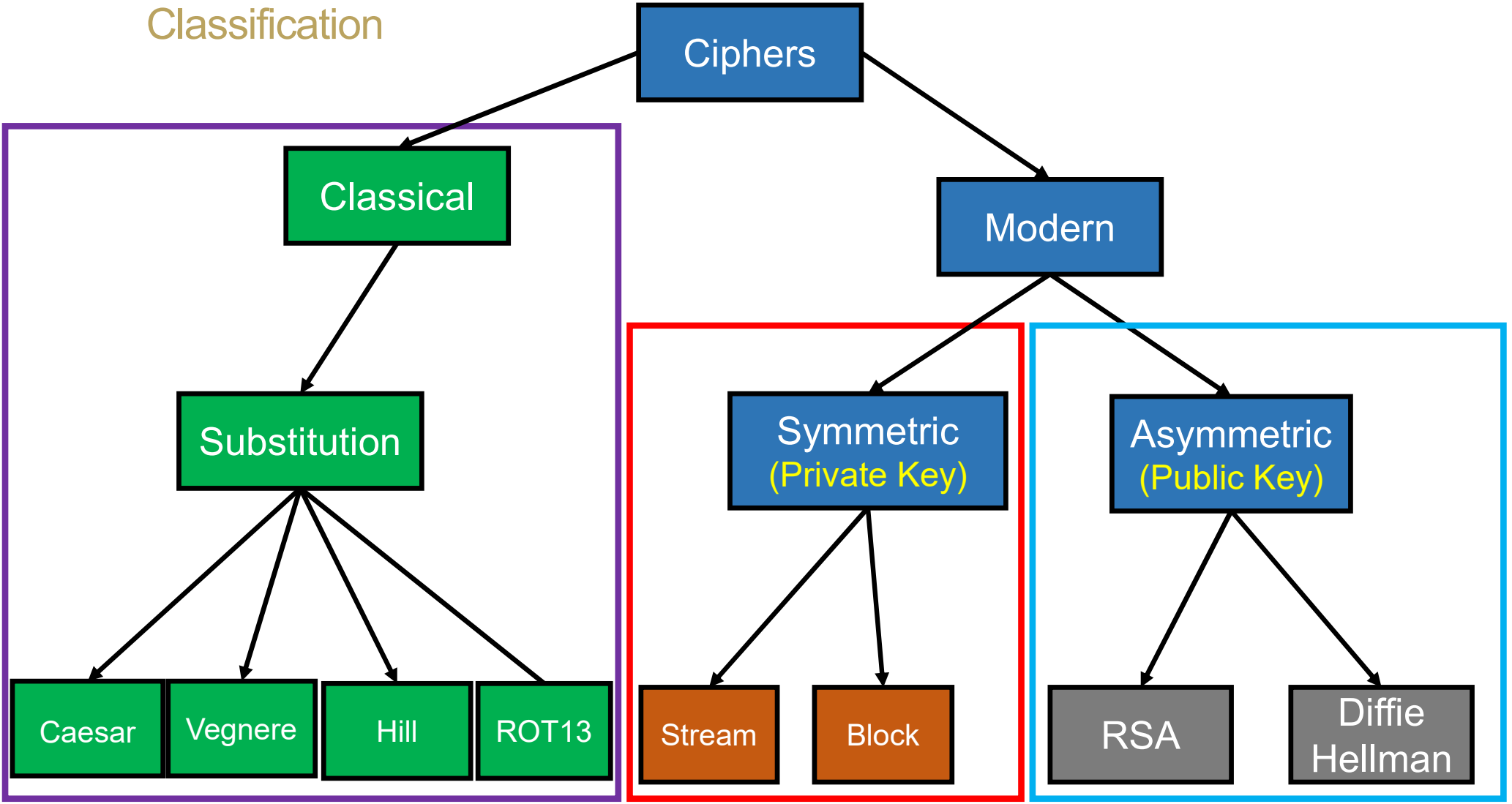
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Introduction

- Cryptography: Definition
- Data Security
- Historical Background
- Symmetric Encryption examples
  - DES
  - AES
- Randomness
- Asymmetric Encryption
  - RSA
  - DH
- Trust in Cryptography

What we have covered

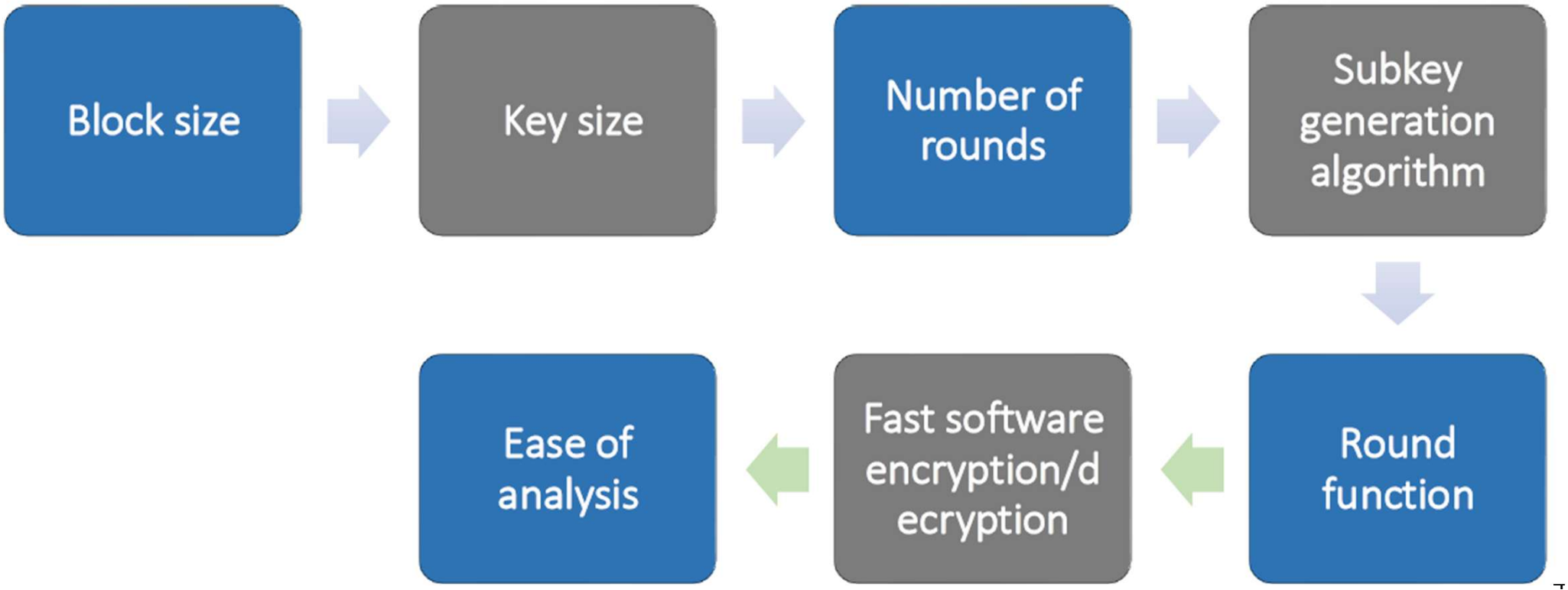
Classification



Symmetric Encryption

Block Cipher

- Processes the input one block of elements at a time
- Produces an output block for each input block
- **Block size:** Larger block sizes mean greater security (all other things being equal) but reduced encryption/decryption speed. A block size of 128 bits is a reasonable trade-off and is nearly universal among recent block cipher designs.



Symmetric Encryption

Data Encryption Standard (DES)

Until recently was the most widely used encryption scheme

- FIPS PUB 46
- Referred to as the Data Encryption Algorithm (DEA)
- Uses 64 bit plaintext block and 56 bit key to produce a 64 bit ciphertext block
- It is the best studied cipher.
- DES is a Feistel Cipher.

Symmetric Encryption

Data Encryption Standard (DES)

Strength concerns:

- Concerns about the algorithm itself
  - DES is the most studied encryption algorithm in existence

  – some detractors but no serious flaws

- Concerns about the use of a 56-bit key
  - The speed of commercial off-the-shelf processors makes this key length woefully inadequate

Symmetric Encryption

Data Encryption Standard (DES)

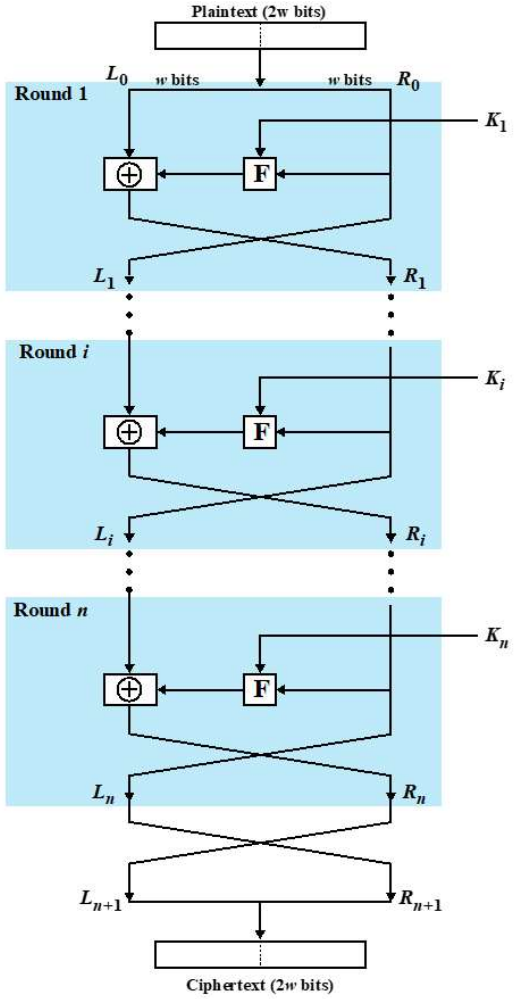
Key size (bits)	Cipher	Number of Alternative Keys	Time Required at 10^9 decryptions/s	Time Required at 10^{13} decryptions/s
56	DES	$2^{56} \approx 7.2 \times 10^{16}$	2^{55} ns = 1.125 years	1 hour
128	AES	$2^{128} \approx 3.4 \times 10^{38}$	2^{127} ns = 5.3×10^{21} years	5.3×10^{17} years
168	Triple DES	$2^{168} \approx 3.7 \times 10^{50}$	2^{167} ns = 5.8×10^{33} years	5.8×10^{29} years
192	AES	$2^{192} \approx 6.3 \times 10^{57}$	2^{191} ns = 9.8×10^{40} years	9.8×10^{36} years
256	AES	$2^{256} \approx 1.2 \times 10^{77}$	2^{255} ns = 1.8×10^{60} years	1.8×10^{56} years

Average Time Required for Exhaustive Key Search

Symmetric Encryption

Feistel Network

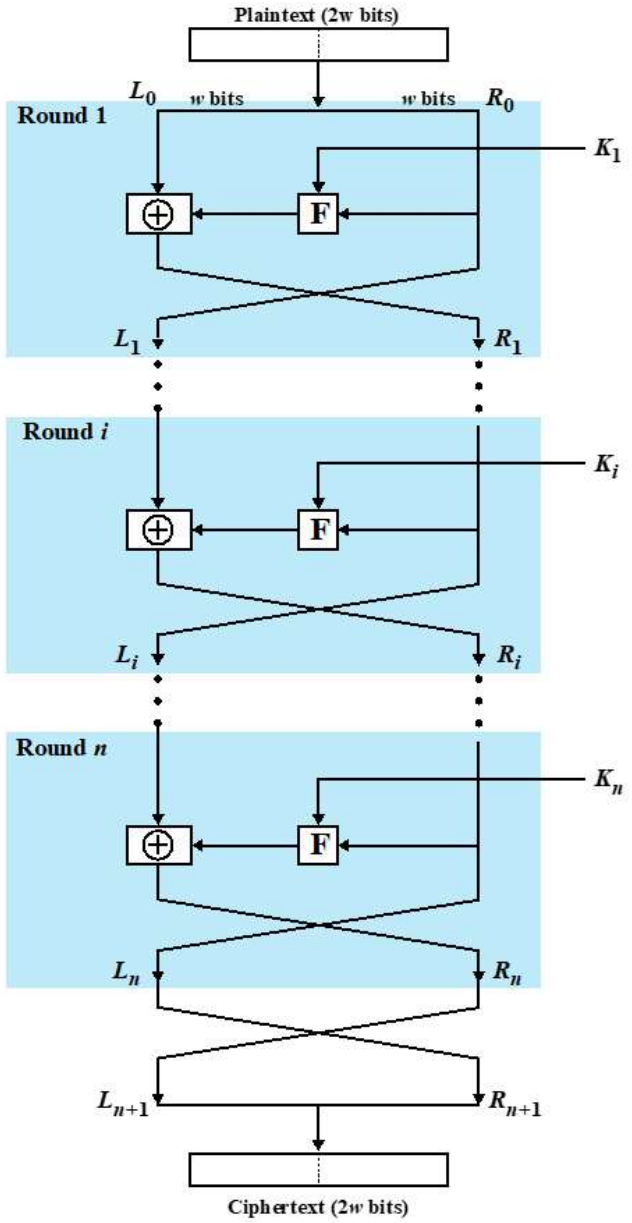
- Many encryption algorithms incorporate a structure first proposed by Horst Feistel of IBM (1973); a **Feistel Network**.
- This includes the DES standard.



Symmetric Encryption

Feistel Network

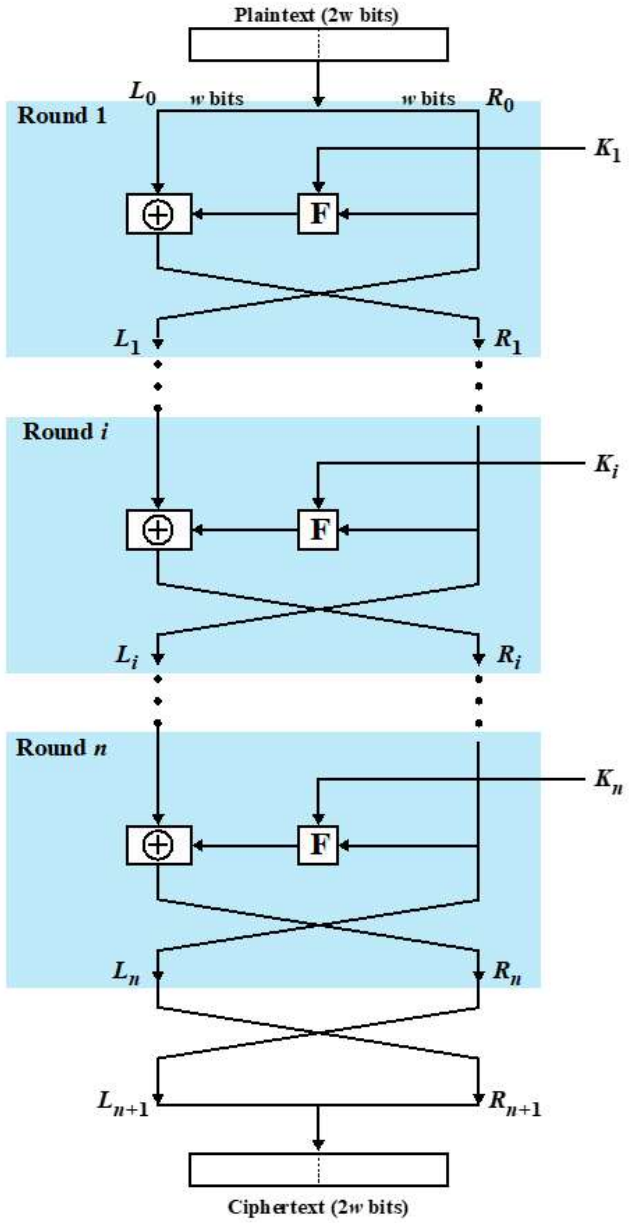
- The inputs to the encryption algorithm are a plaintext block of length 2w bits and a key K .
- The plaintext block is divided into two halves, L^0 and R^0.
- The two halves of the data pass through n rounds of processing and then combine to produce the ciphertext block.
- Each round i has as inputs L^{i-1} and R^{i-1} , derived from the previous round, as well as a subkey K^i, derived from the overall K .
- In general, the subkeys K^i are different from K and from each other and are generated from the key by a subkey generation algorithm.



Symmetric Encryption

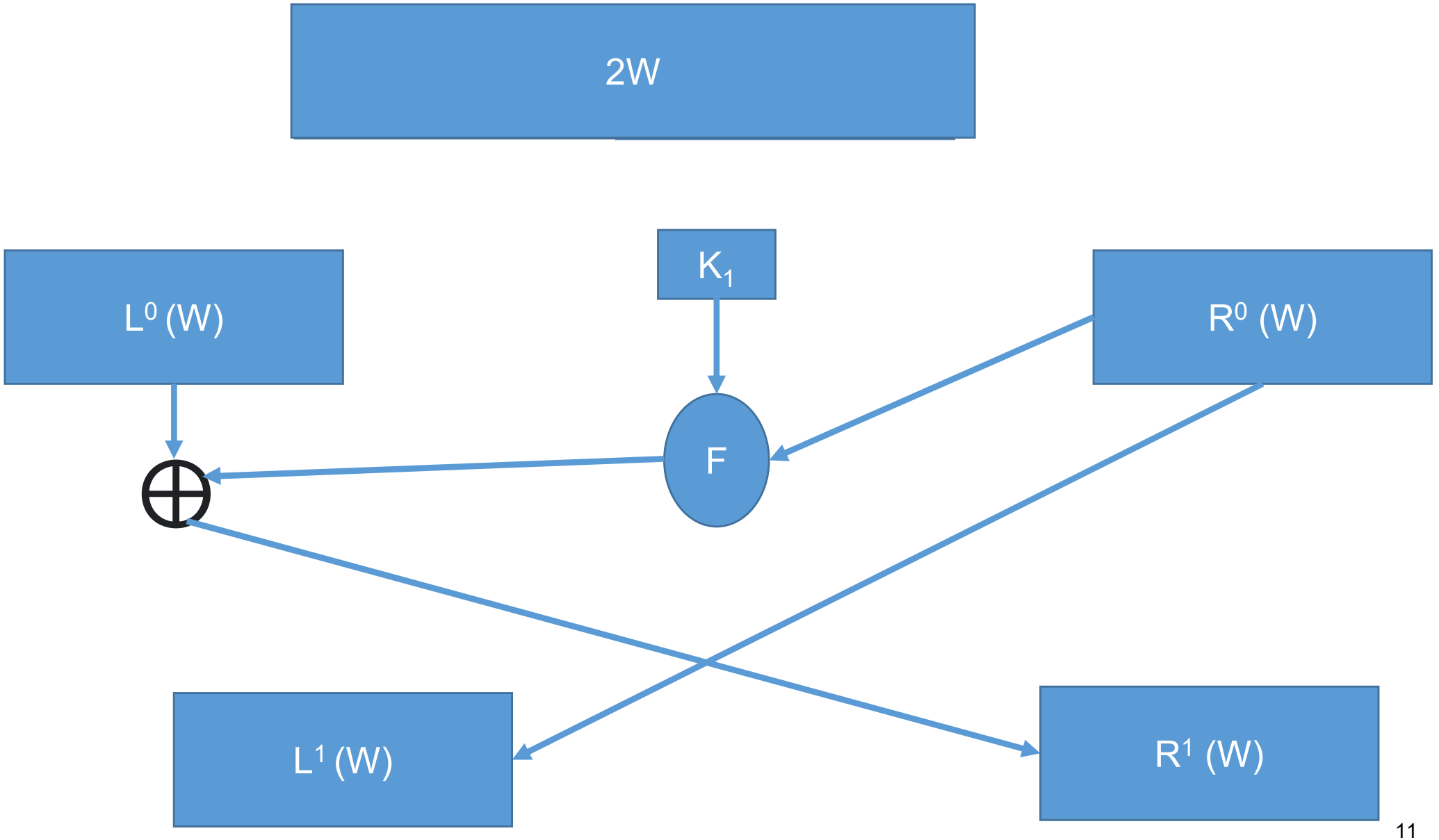
Feistel Network

- All rounds have the same structure. A substitution is performed on the left half of the data.
- This is done by applying a round function F to the right half of the data and then taking the exclusive-OR (XOR) of the output of that function and the left half of the data.
- The round function has the same general structure for each round but is parameterized by the round subkey K^i .
- Following this substitution, a permutation is performed that consists of the interchange of the two halves of the data.



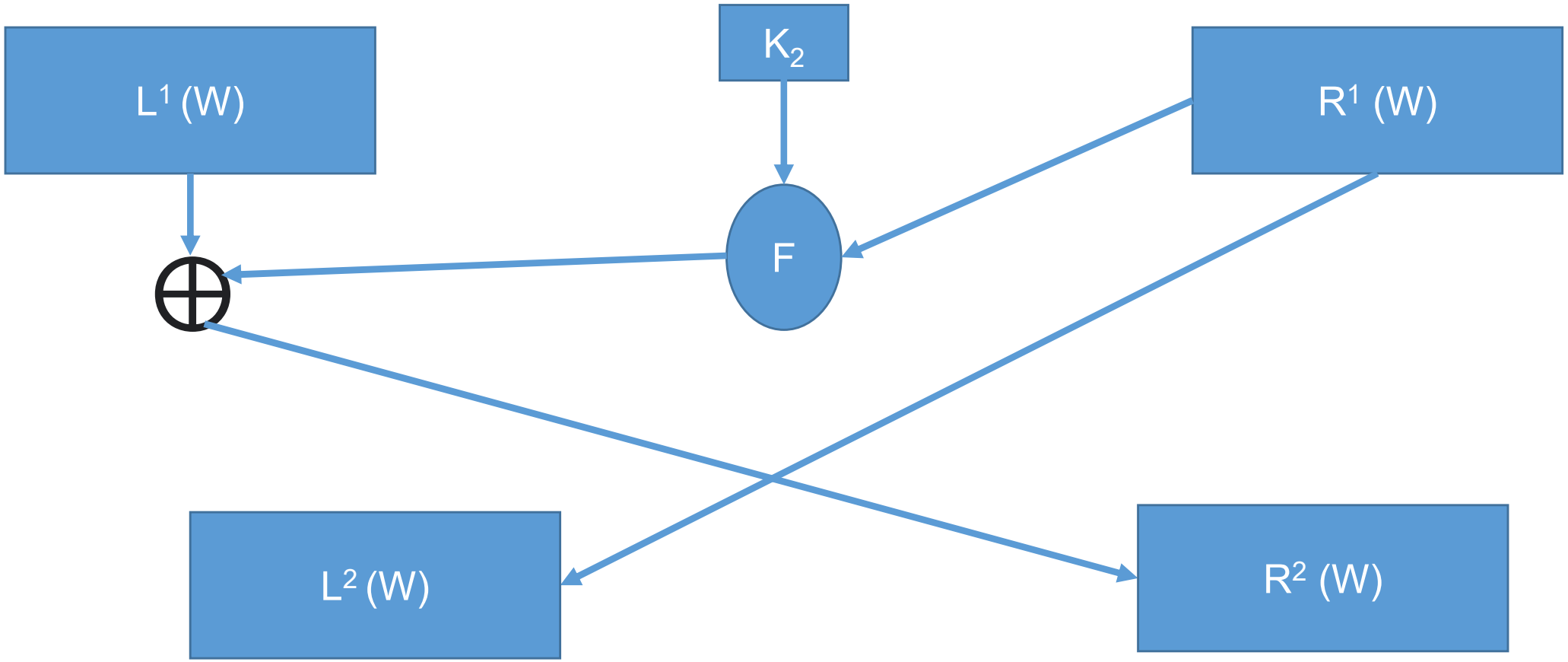
Symmetric Encryption

Feistel Network



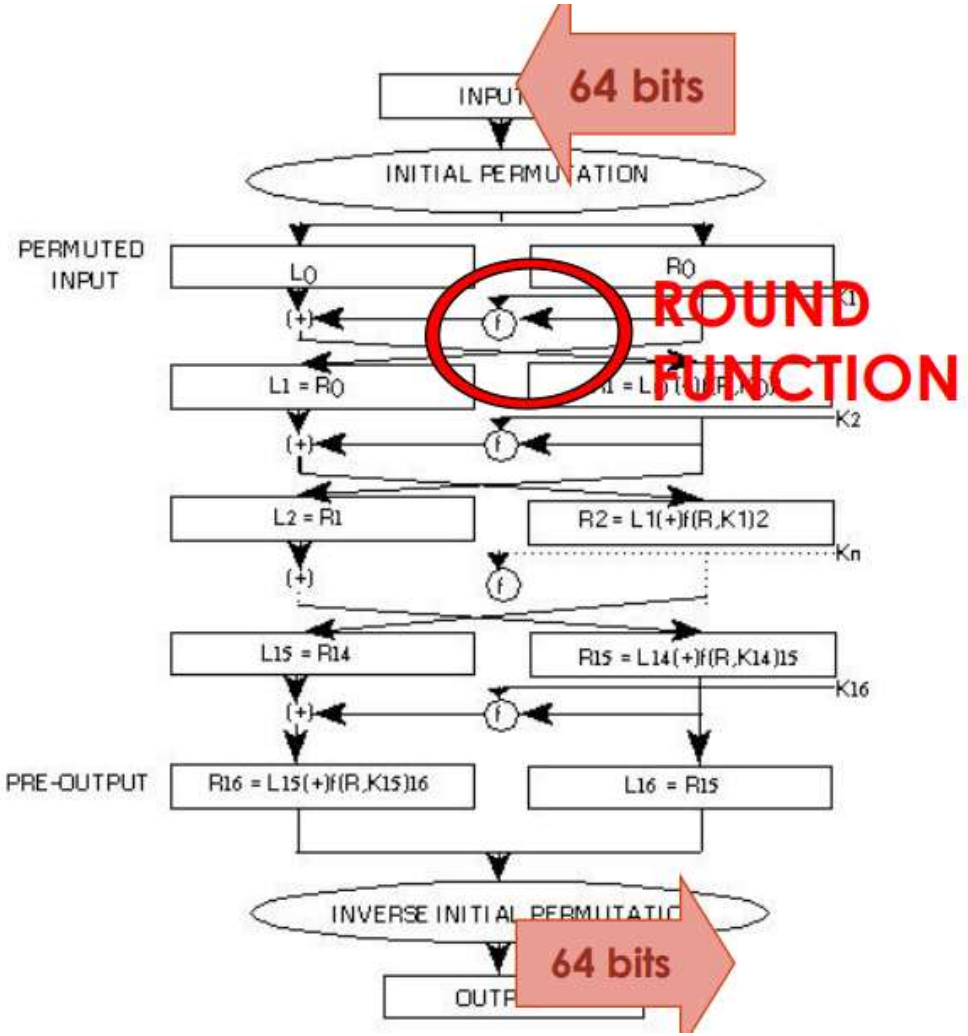
Symmetric Encryption

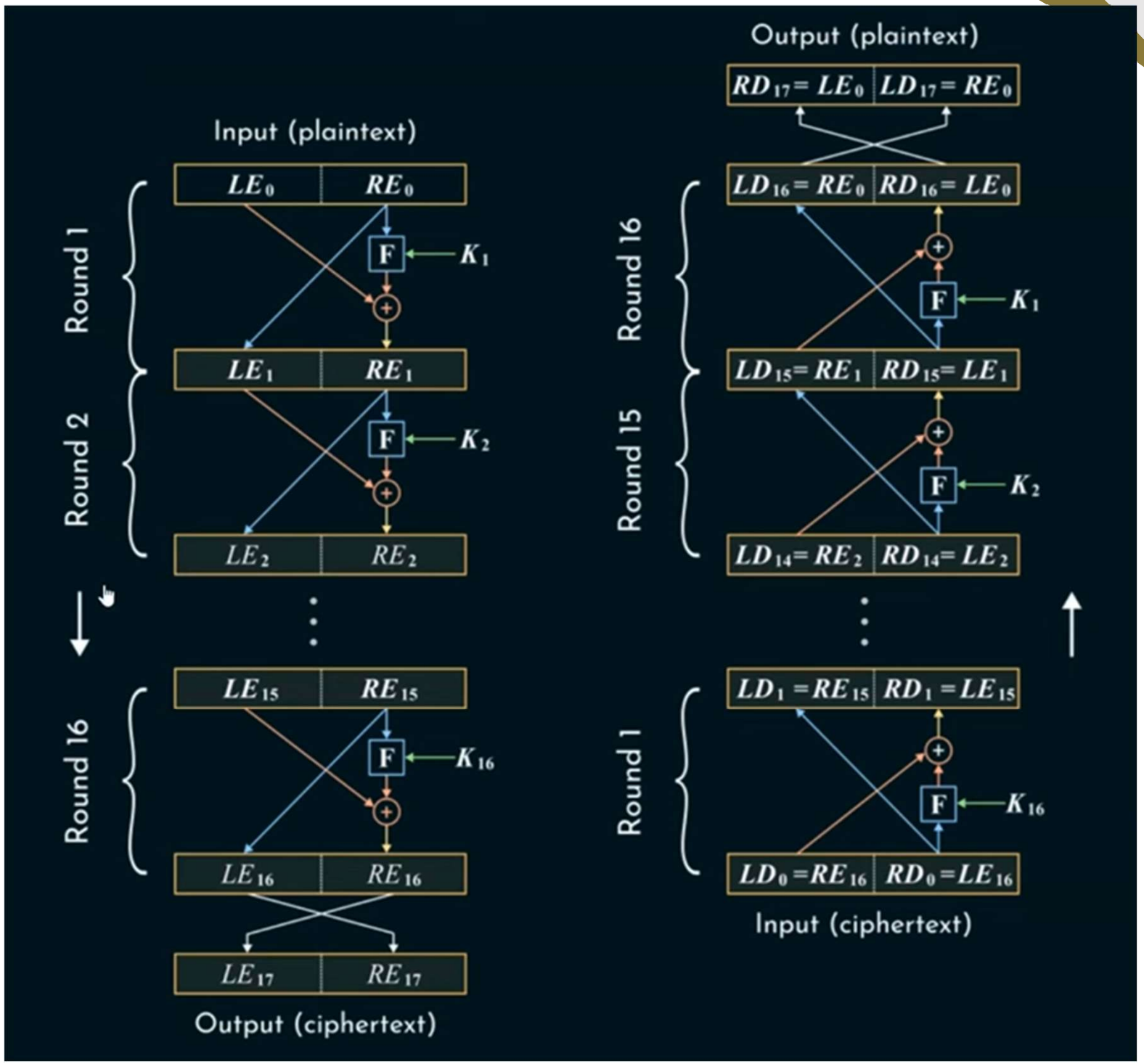
Feistel Network



DES

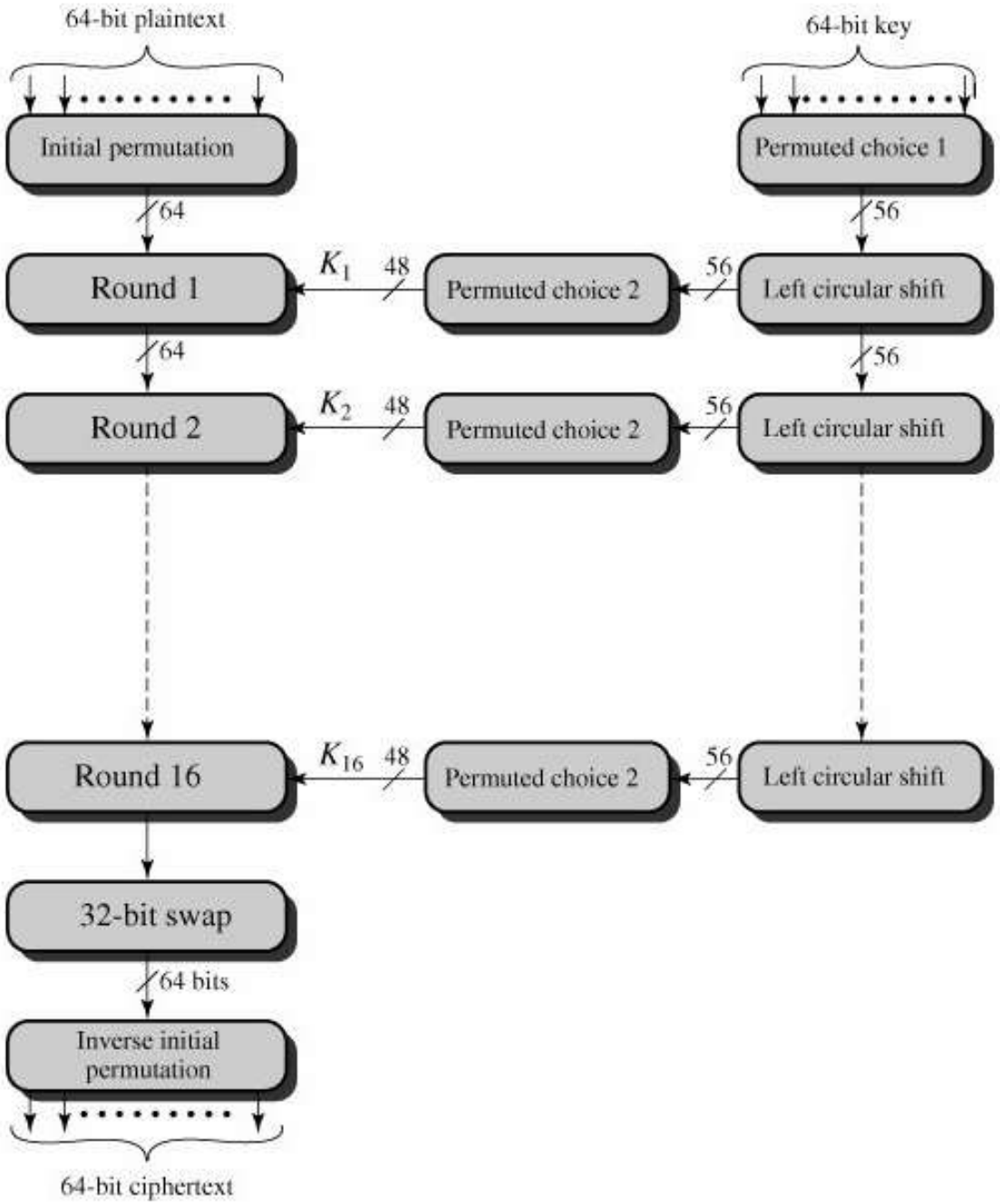
- 16 Rounds.
- In each round we apply:
  - Expansion Box.
  - Substitution Box.
  - XOR with the round key.





Screenshot from Youtube video

Figure 3.4. General Depiction of DES Encryption Algorithm



Initial Permutation

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30	31	32
33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48
49	50	51	52	53	54	55	56
57	58	59	60	61	62	63	64

Initial Permutation

64-Bit Plain Text Box

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30	31	32
33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48
49	50	51	52	53	54	55	56
57	58	59	60	61	62	63	64

Permutation



64-Bit Permutated Output

58	50	42	34	26	18	10	2
60	52	44	36	28	22	12	4
62	54	46	38	30	22	14	6
64	56	48	40	32	24	16	8
57	49	41	32	25	17	9	1
59	51	43	33	27	19	11	3
61	53	45	35	29	21	13	5
63	55	47	37	31	23	15	7

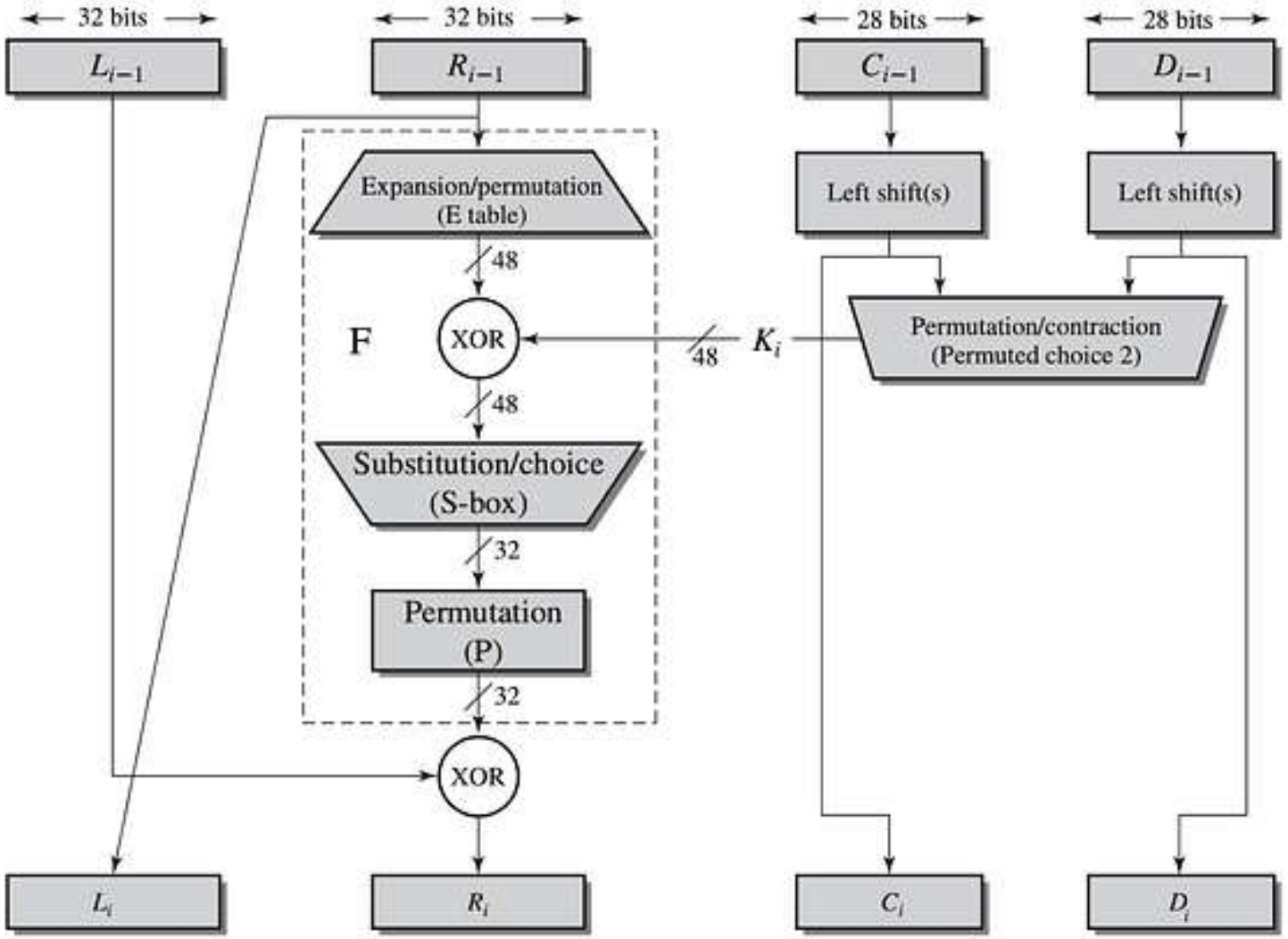
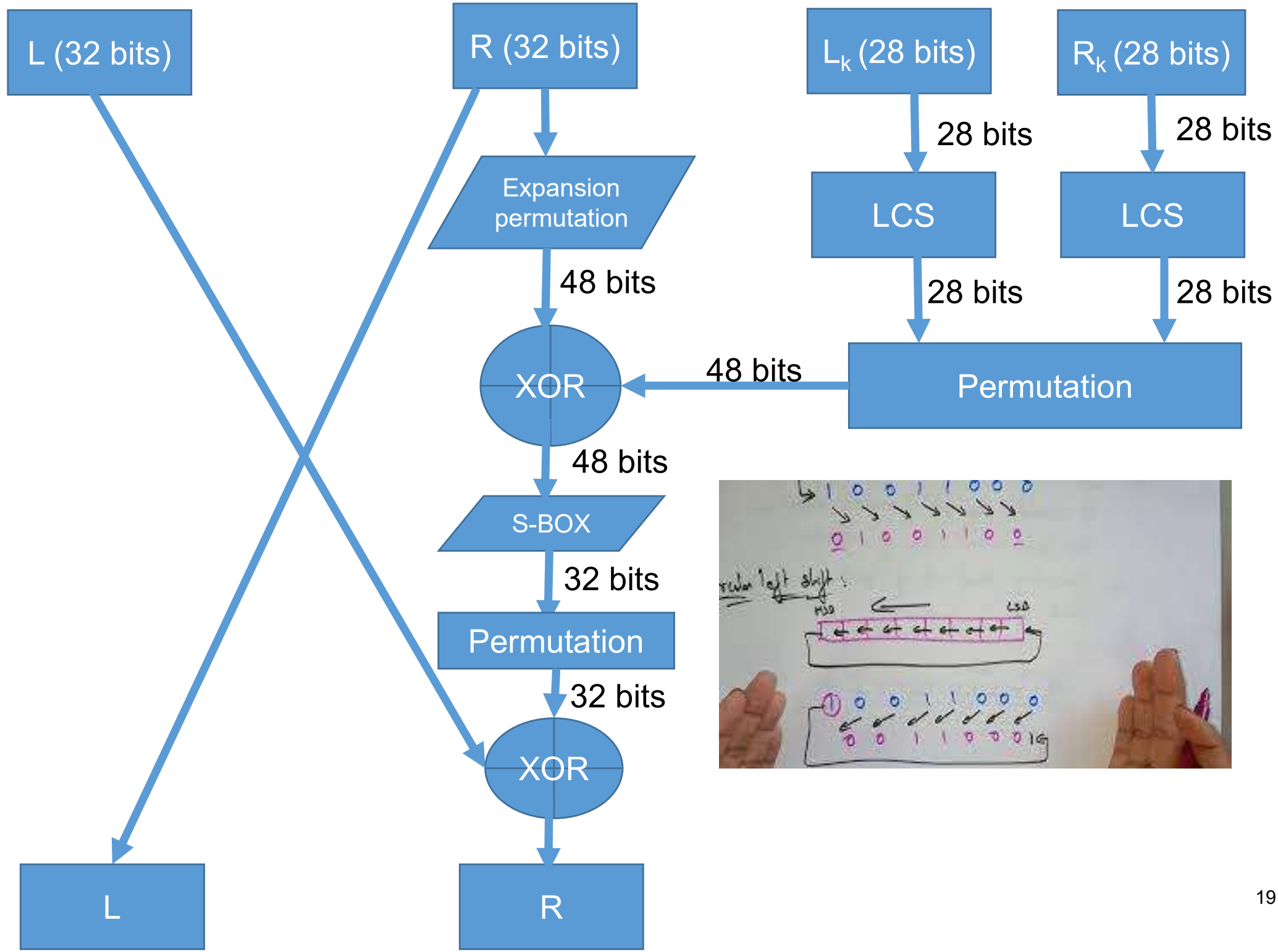


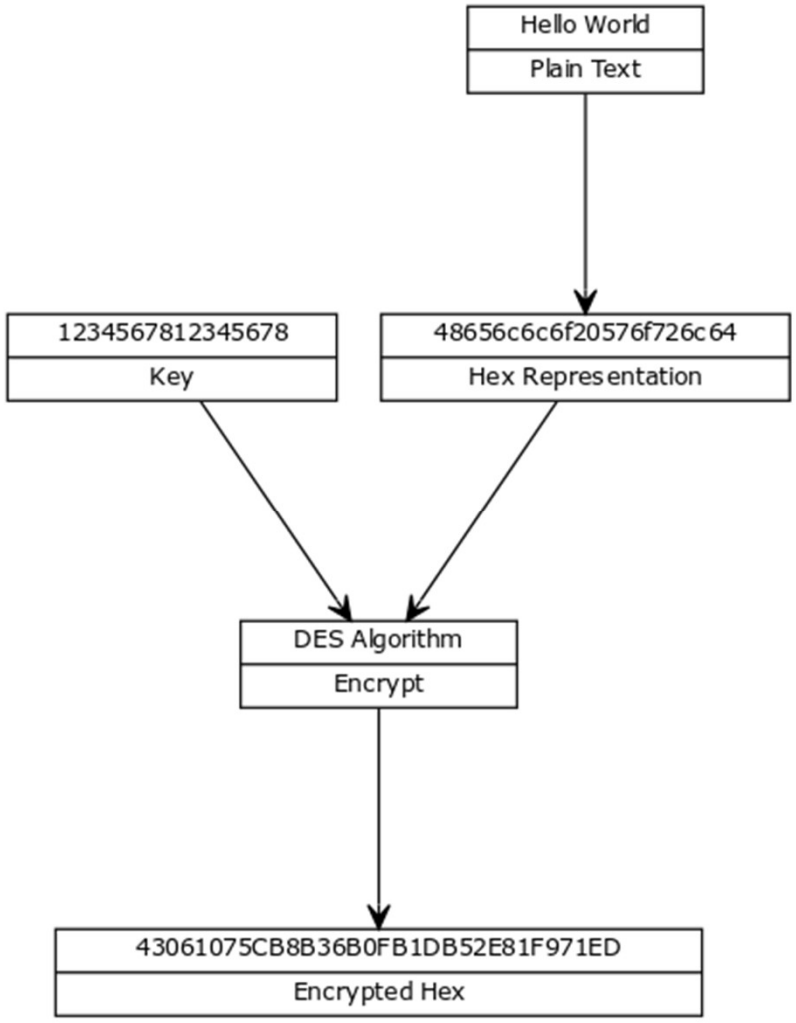
Figure 3.6 Single Round of DES Algorithm

Figure From TextBook W. Stallings, Cryptography and Network Security: Principles and Practice


L (32 bits)
R (32 bits)
L_k (28 bits)
R_k (28 bits)
28 bits
28 bits
Expansion permutation
LCS
LCS
48 bits
28 bits
28 bits
XOR
48 bits
Permutation
48 bits
S-BOX
32 bits
Permutation
32 bits
XOR
L
R

Symmetric Encryption

Data Encryption Standard (DES)

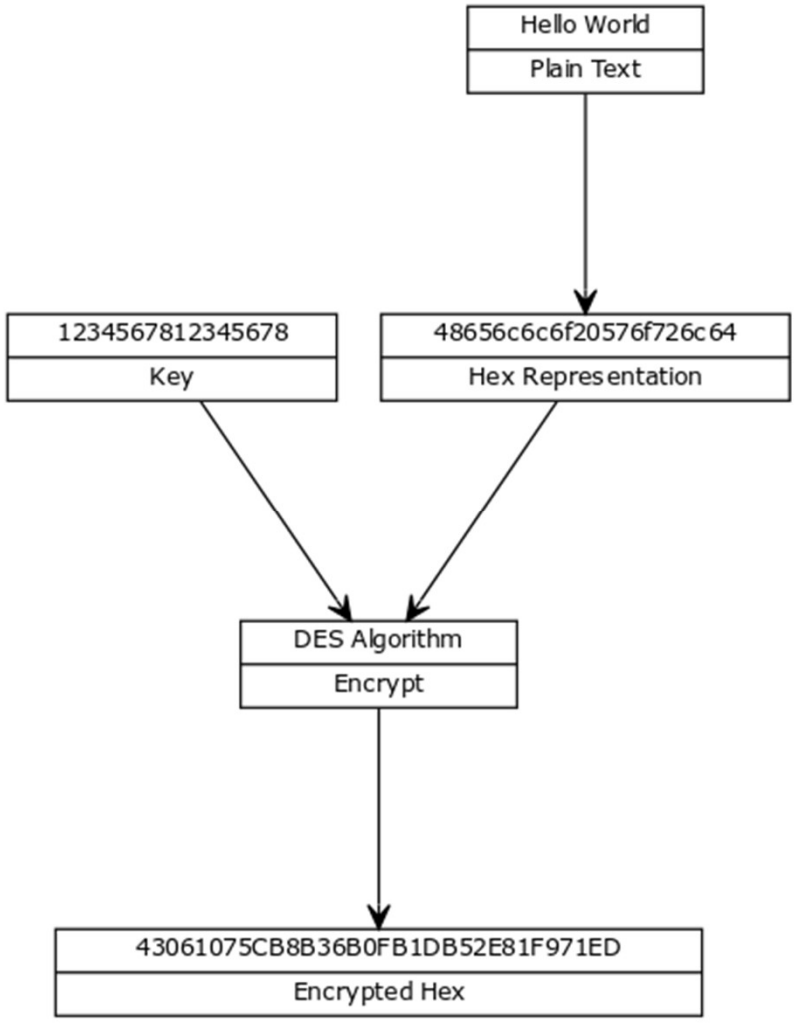


Variable Changes
Guess the outcome if we change the following

1. **Plaintext:** Hello world
2. **Key Change:** 00000000*

Symmetric Encryption

Data Encryption Standard (DES)



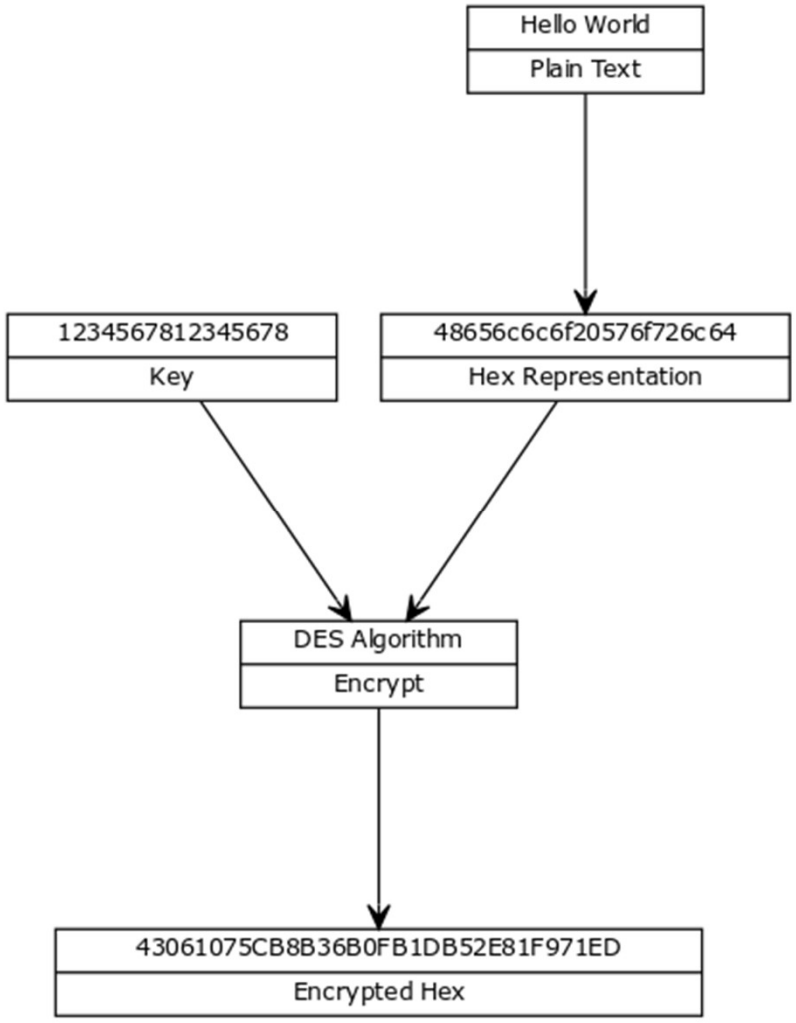
Variable Changes

Plaintext: Hello world
Hex: 48656c6c6f20776f726c64
Output:
80508BC5A5F89985FB1DB52E8
1F971ED

Key Change: 0000000000000000
Output:
EADE4DB3AFA20320011F38638
1D8E123

Symmetric Encryption

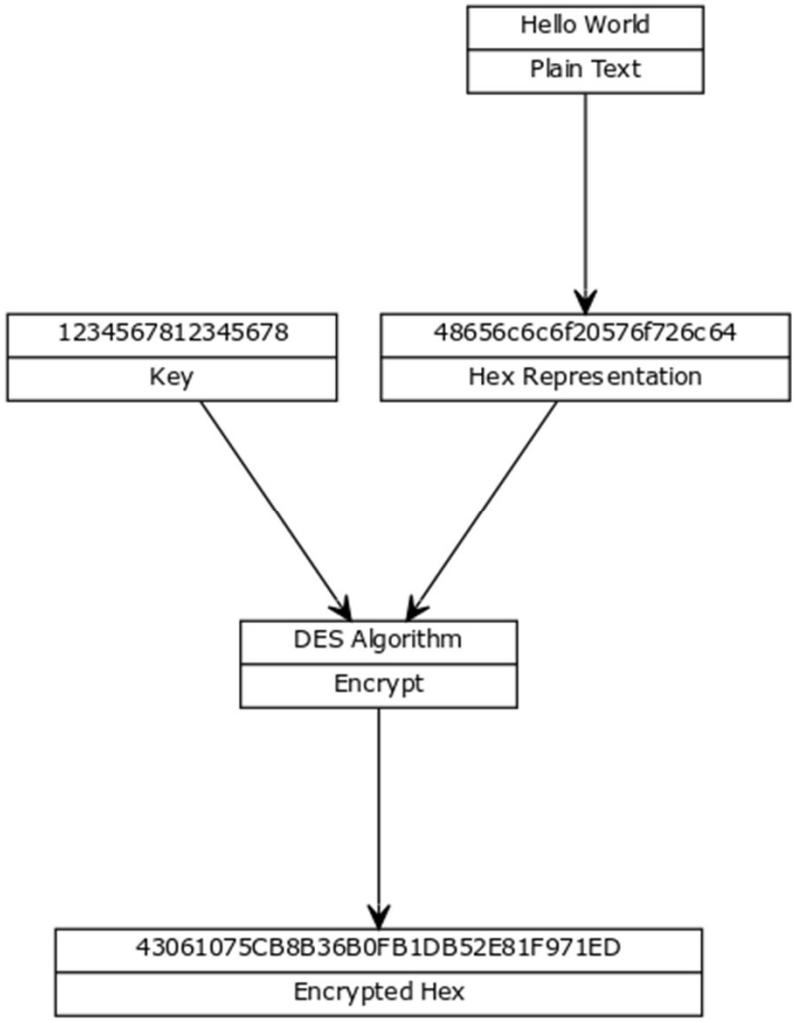
Data Encryption Standard (DES)



Why is the Encrypted Hex much longer?

Symmetric Encryption

Data Encryption Standard (DES)



Why is the Encrypted Hex much longer?

Why is this important?

Symmetric Encryption

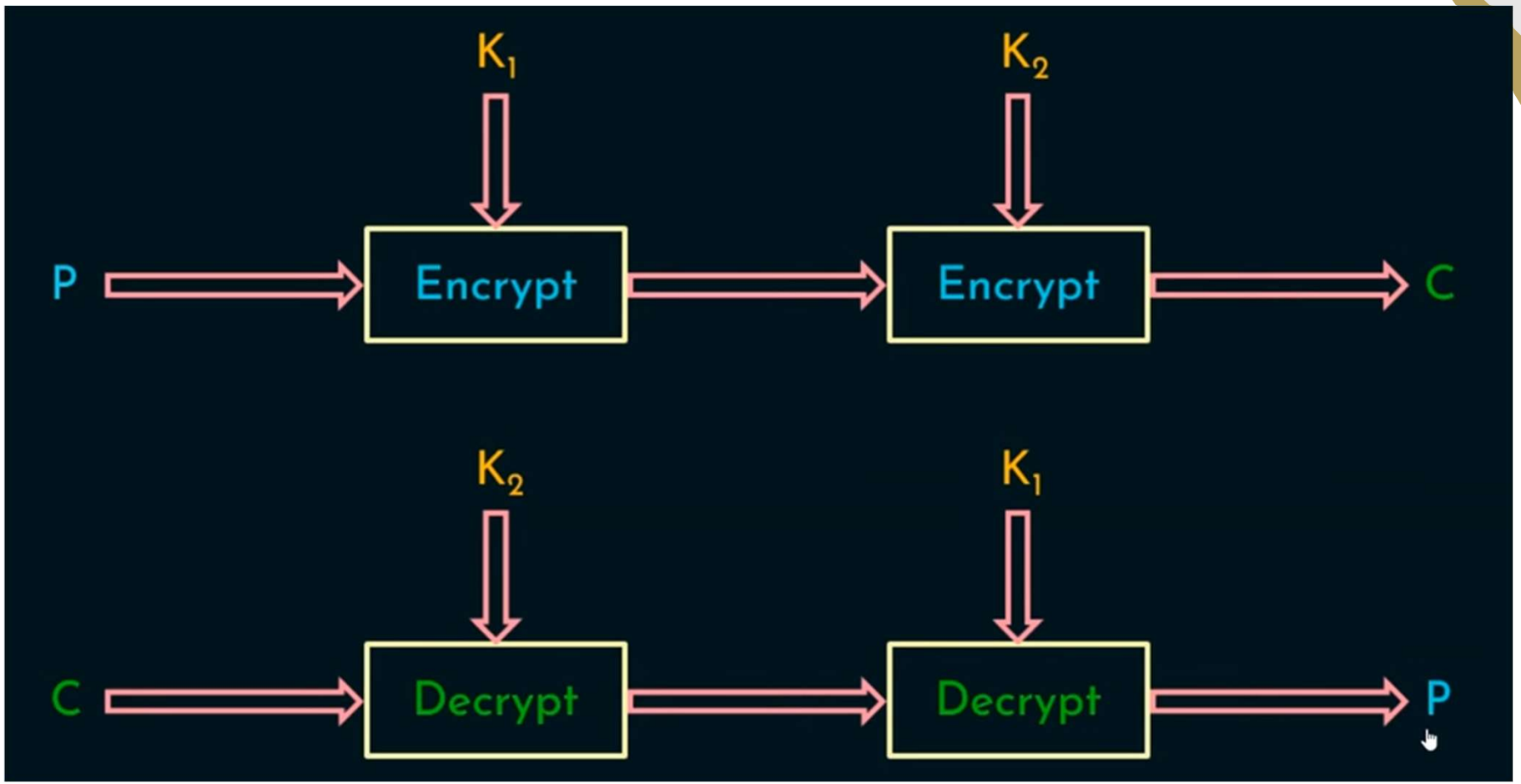
Data Encryption Standard (DES)

Brute-force attacks

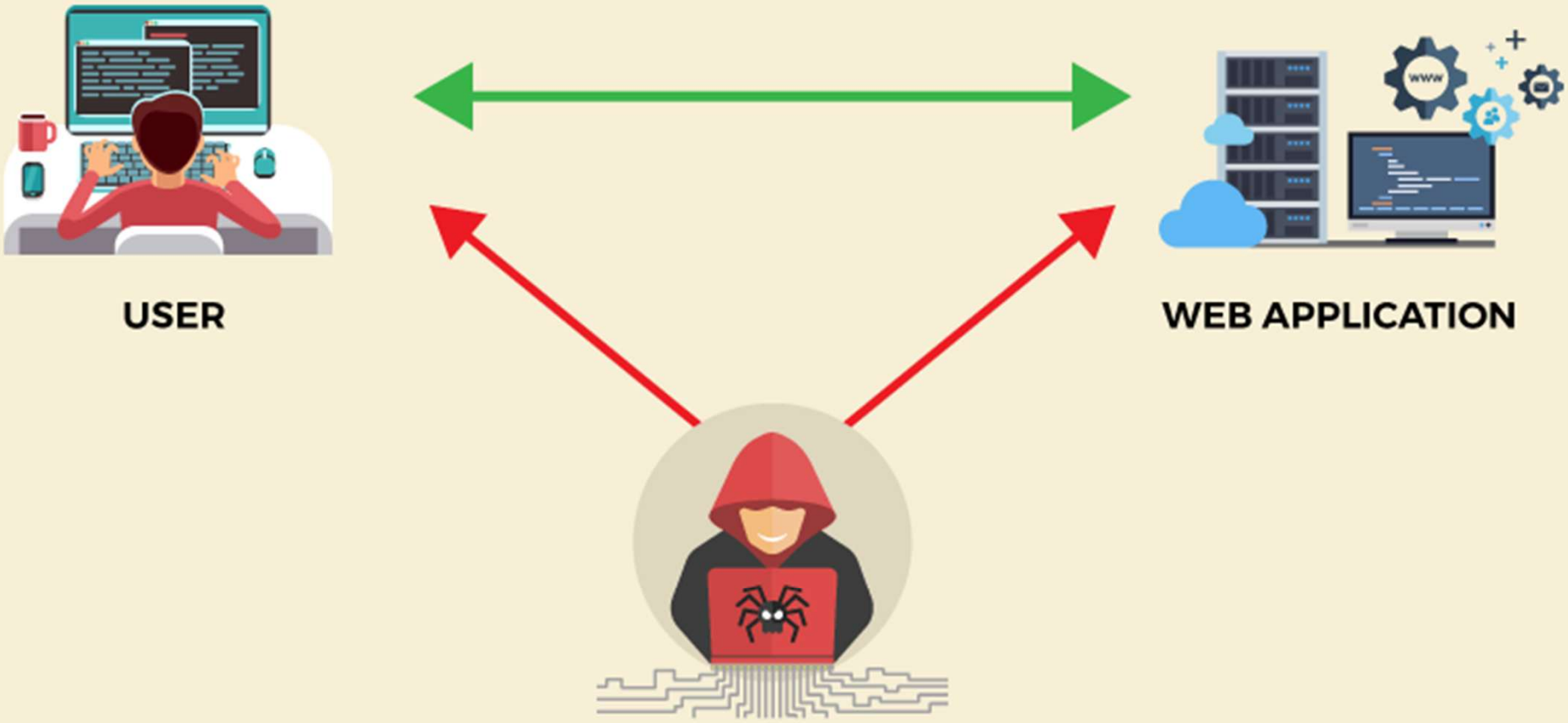
- Try all possible keys on some ciphertext until an intelligible translation into plaintext is obtained
- On average half of all possible keys must be tried to achieve success
- Number of Keys are dictated by the key size.

Symmetric Encryption

Double DES (2DES)



HOW MAN-IN-THE-MIDDLE ATTACK WORKS



Symmetric Encryption

Triple DES (3DES)

- Repeats basic DES algorithm three times using either two or three unique keys

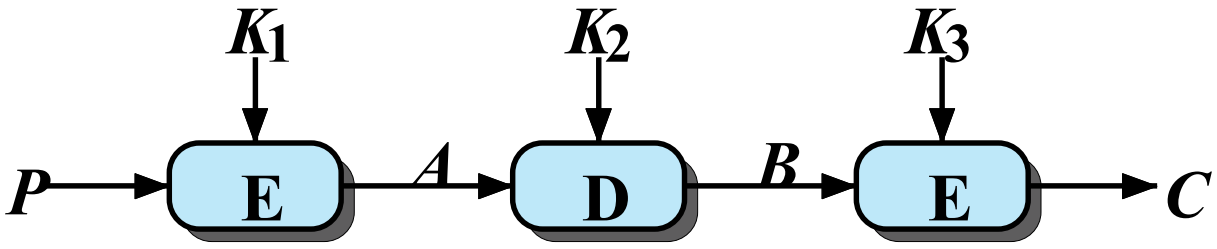
- First standardized for use in financial applications in ANSI standard X9.17 in 1985

Attractions:
- 168-bit key length overcomes the vulnerability to brute-force attack of DES
- Underlying encryption algorithm is the same as in DES

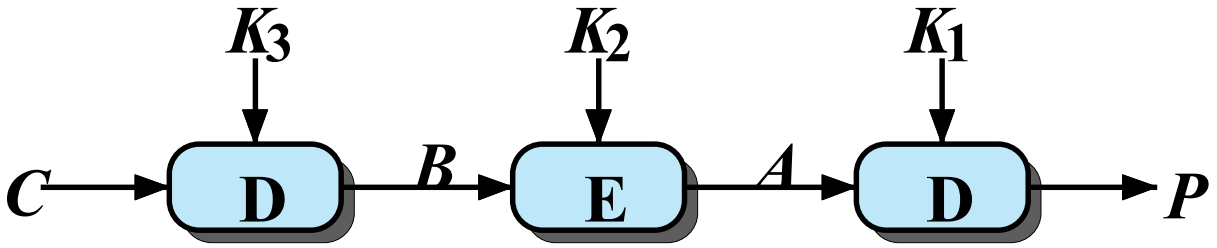
Drawbacks:
- Algorithm is sluggish in software
- Uses a 64-bit block size

Symmetric Encryption

Triple DES (3DES)



(a) Encryption



(b) Decryption

Symmetric Encryption

Advanced Encryption Standard (AES)

Needed a replacement for 3DES

3DES was not reasonable for long term use

NIST called for proposals for a new AES in 1997

- **Should have a security strength equal to or better than 3DES**
- **Significantly improved efficiency**
- **Symmetric block cipher**
- **128 bit data and 128/192/256 bit keys**

Selected Rijndael in November 2001

Published as FIPS 197

AES

An **iterative** rather than **Feistel** cipher

- Processes data as block of 4 columns of 4 bytes
- Operates on entire data block in every round

Designed to be:

- Resistant against known attacks
- Speed and code compactness on many CPUs
- Design simplicity

Data block viewed as 4-by-4 table of bytes

Such a table is called the **current state**

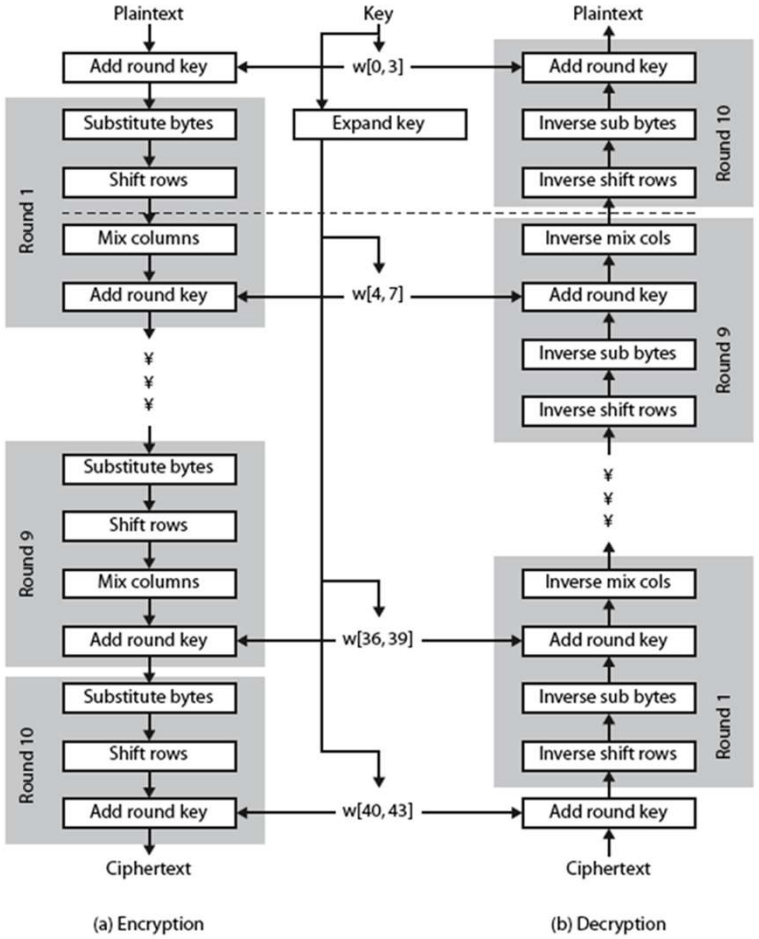
Key is expanded to array of words

Has 10 rounds in which state the following transformations (called `layers'):

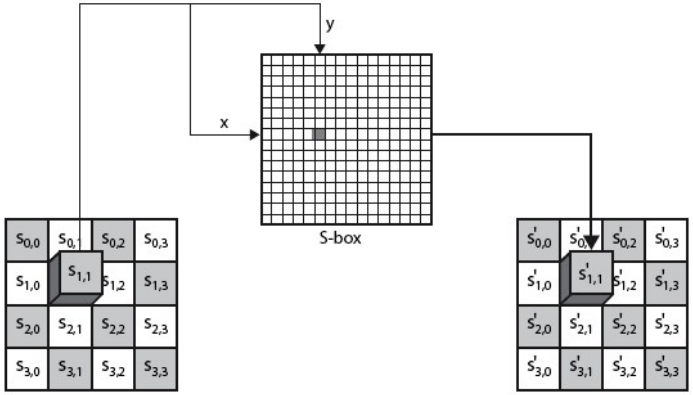
- BS- byte substitution (1 S-box used on every byte)
- SR- shift rows (permute bytes between groups/columns)
- MC- mix columns (uses matrix multiplication in GF(256))
- ARK- add round key (XOR state with round key)

First and last round are a little different

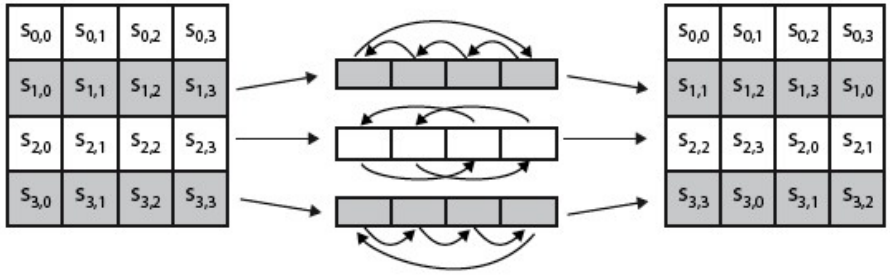
AES



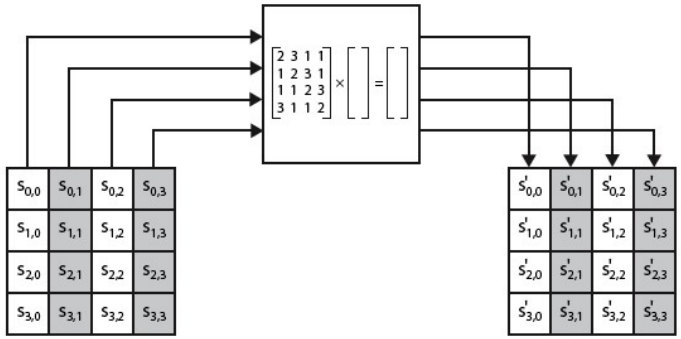
BS



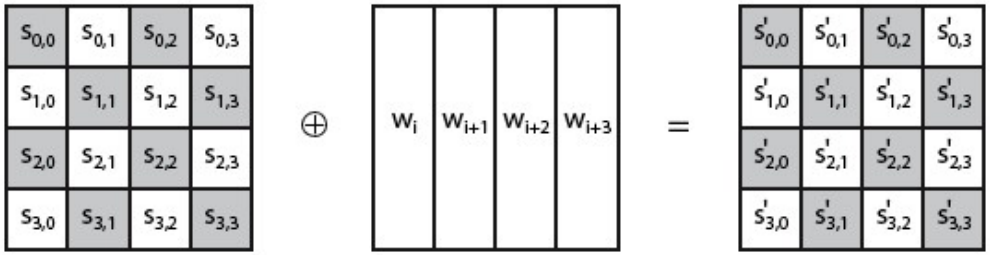
SR

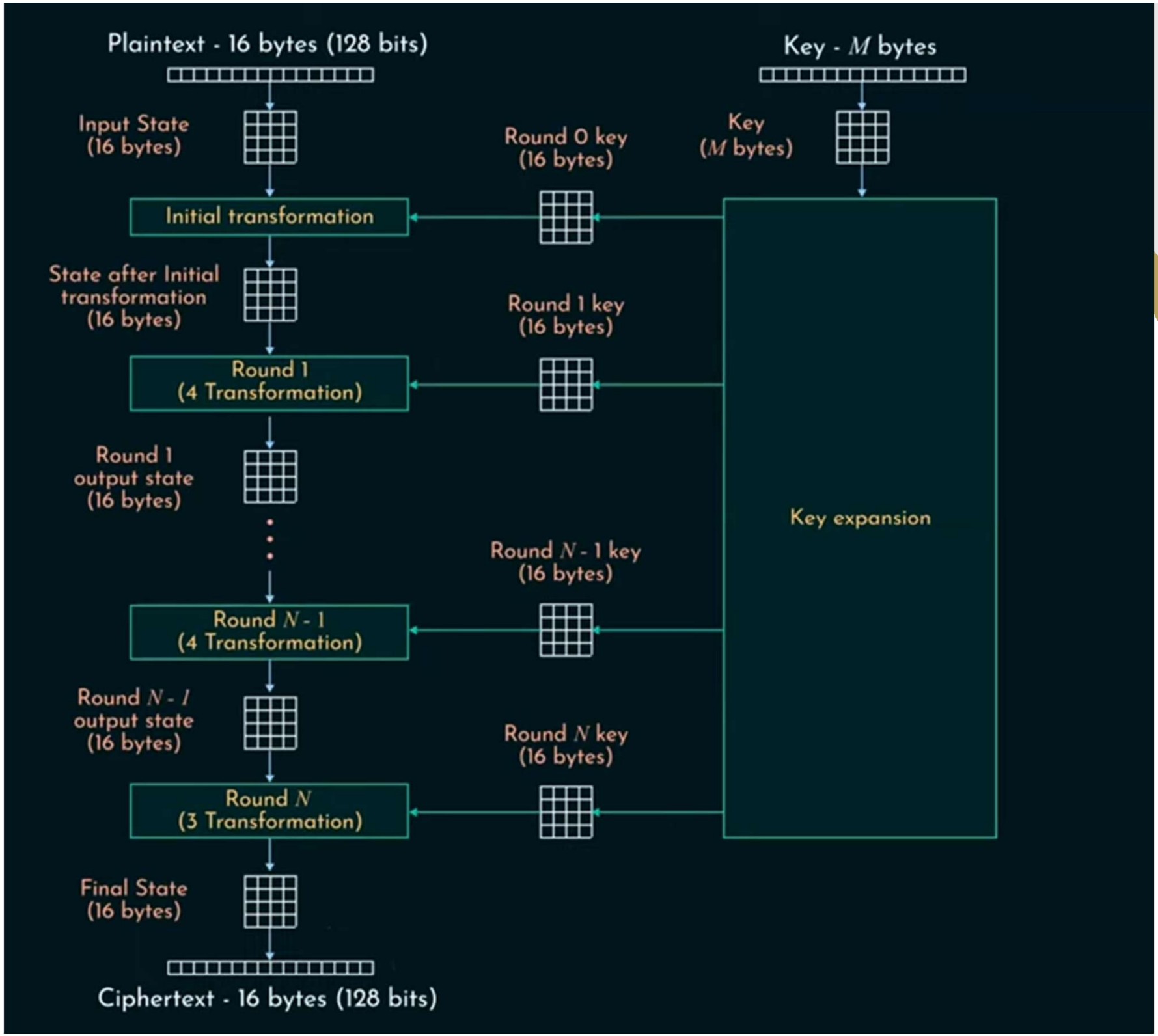


MC



ARK




Plaintext - 16 bytes (128 bits)
Key - M bytes
Input State
(16 bytes)
Round 0 key
(16 bytes)
Key
(M bytes)
Initial transformation
State after Initial
transformation
(16 bytes)
Round 1 key
(16 bytes)
Round 1
(4 Transformation)
Round 1
output state
(16 bytes)
Key expansion
Round N - 1 key
(16 bytes)
Round N - 1
(4 Transformation)
Round N - 1
output state
(16 bytes)
Round N key
(16 bytes)
Round N
(3 Transformation)
Final State
(16 bytes)
Ciphertext - 16 bytes (128 bits)

AES Parameters

No of rounds	Key size (bits)
10	128
12	192
14	256

	AES-128	AES-192	AES-256
Key Size	128	192	256
Plaintext Size	128	128	128
Number of Rounds	10	12	14
Round Key Size	128	128	128

Random Numbers

Use

Random numbers are essential to many aspects of cryptography, including:

- Keys for public-key algorithms
- Stream key for symmetric stream cipher
- Symmetric key for use as a temporary session key or in creating a digital envelope
- Handshaking to prevent replay attacks
- Session key

Random Numbers
Criteria

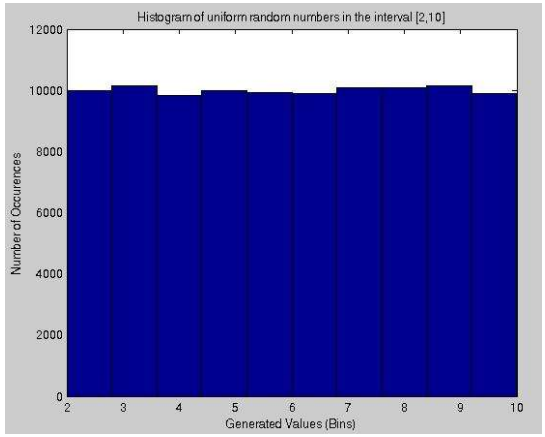
Criteria for a good Random Number Generator

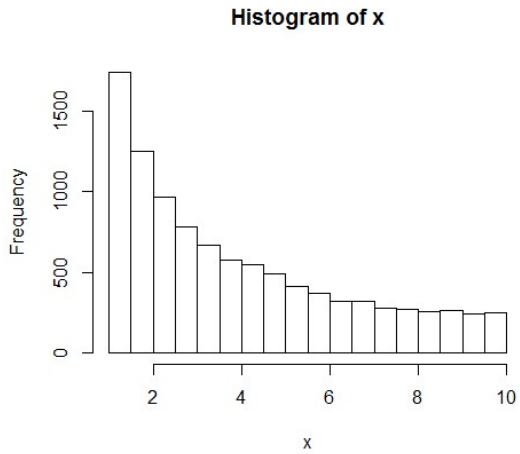
Uniform distribution of Random Numbers:

- Frequency of occurrence of each of the numbers should be approximately the same

Independence of Random Numbers:

- No one value in the sequence can be inferred from the others





Random Numbers

Criteria

Criteria for a good Random Number Generator

Unpredictability

- Each number is statistically independent of other numbers in the sequence
- Opponent should not be able to predict future elements of the sequence on the basis of earlier elements

Random Numbers

Random vs Pseudorandom

Cryptographic applications typically make use of algorithmic techniques for random number generation

- Algorithms are deterministic and therefore produce sequences of numbers that are not statistically random

Pseudorandom numbers are:

- Sequences produced that satisfy statistical randomness tests
- Likely to be predictable

True random number generator (TRNG):

- Uses a nondeterministic source to produce randomness
- Most operate by measuring unpredictable natural processes
  - e.g. radiation, gas discharge, leaky capacitors
- Increasingly provided on modern processors

Public-Key Cryptography

- Probably most significant advancement in the 3000 year history of cryptography
- Uses two keys – a public & a private key
- Asymmetric since parties are not equal
- Uses clever application of number theoretic concepts to function
- complements rather than replaces private key crypto

Why Public-key

- Developed to address two key issues:
  - **key distribution** – how to have secure communications in general without having to trust a KDC with your key
  - **digital signatures** – how to verify a message comes intact from the claimed sender
- Public invention due to Whitfield Diffie & Martin Hellman at Stanford Uni in 1976
  - known earlier in classified community

Public-Key Cryptography

